Status	Finished
Started	Thursday, 29 October 2026, 8:41 PM
Completed	Thursday, 29 October 2026, 8:45 PM
Duration	4 mins 20 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the largest element to its correct position
c. Bubbling up the smallest element
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. Each part is searched for a specific element
c. The parts are merged without sorting
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It divides the list into sublists
c. It bubbles up the largest element in each pass
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Heap Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120
b. 80,150,200,470,40,90,300,400,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. order()
- d. arrange()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. All of the above
- c. Selection
- d. Searching

The correct answer is: All of the above

Question **12**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Sorting data in a non-sequential manner
c. Bubbling up the largest element to its correct position with each pass
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Complexity

c. Searching

d. Merging

The correct answer is:
Complexity